Ji et al. 2020 — Cutaneous Squamous Cell Carcinoma

CCCA meta-atlas · cell-state manifold & embryonic-program analysis

Skin · 10x · 10,523 malignant cells (8 / 10 patients) · geometry: BRANCHING (hub-and-spoke)

GO / NO-GO GATES

Gate 1 — mixing + continuity — GO

Cell-cycle regressed out (de-novo S/G2M scores), Harmony on sample (theta=2), n_pcs=20. Embedding is sample-integrated (all 8 patients interleave; 30-NN cross-sample fraction 0.63 vs 0.78 max) and one continuous body (single radius-graph component). Without CC regression a spurious cycling axis dominates a lobe — the confound the prior linear read never removed.

Gate 2 — module geography — GO

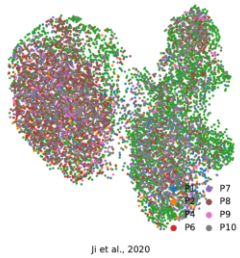
Basal Gast/pSom hub diverges into a Gast-leaning differentiation arm (KRT1/10; Gast +0.00 vs hub -0.04) and a pSom-leaning TSK / leading-edge arm (pSom +0.06). Crest is a terminal readout enriched on the pSom (TSK) arm and depleted at the differentiation tip — uncoupled from the Gast/pSom axis (acceptable).

Gate 3 — hub cross-repression — GO

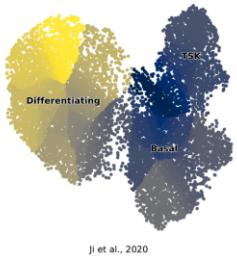
On the basal hub (n=3,756), Gast-loading and pSom-loading genes anti-correlate: mutual r = -0.38 (Puram anchor -0.49; Filbin branching -0.34). Competitive cross-repression at the branch point.

STEP FIGURES

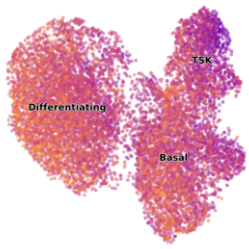
01 Samples (integration)



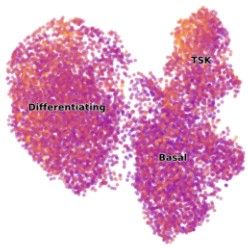
02 Manifold (pseudotime)



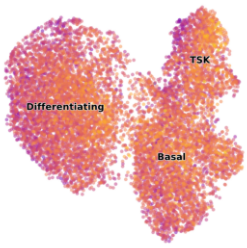
03 Gast



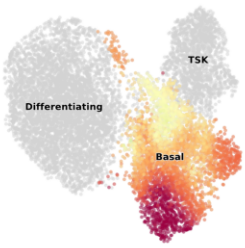
03 pSom



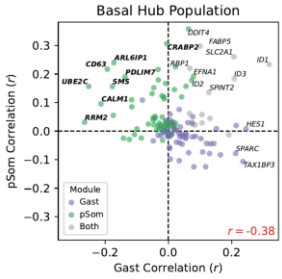
03 Crest



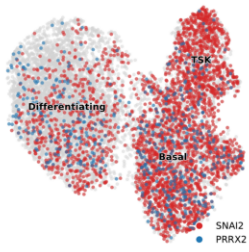
04 Module correlation



05 Hub r = -0.38



06 EMT-TF (SNAI2/PRRX2)



KEY PARAMETER CHOICES

- Threshold: keep samples >=103 (gap-finding: 40->268, ~6.7x) — 8/10 patients, 10,598 -> 10,531 -> 10,523 (island removal) cells.
- Cell cycle: de-novo S/G2M scores regressed out of the over-dispersed matrix (the clean-embedding step).
- Embedding: 1,443 over-dispersed genes, PCA 50 -> Harmony(sample, theta=2), n_pcs=20, UMAP spread=1.3.
- Trajectory: scFates PPT tree (Nodes=80, lambda=600, sigma=0.4) on X_umap; root = basal-max fork.
- Hub: Basal progenitor cluster (branch-point / root convergence zone); subtypes by markers + Ji et al. calls.

VERDICT

Pattern REPRODUCES (branching). A basal progenitor hub diverges into a keratinocyte-differentiation arm (Gast) and a TSK / leading-edge EMT arm (pSom) with hub cross-repression r = -0.38; Crest is a terminal readout on the pSom arm. G1 GO / G2 GO / G3 GO — the third branching case in the atlas.